

GULLFAX PLATFORM, Norway

WMO: 013755

Lat: 61.033N

Lon: 2.033E

Elev: 33

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.6	32.4	17.2	13.1	34.4	19.4	14.6	35.4	69.7	39.3	64.8	40.3	30.1	330	1.171

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	64.8	59.2	62.4	58.5	60.5	57.4	61.0	63.7	59.3	61.5	57.7	59.6	18.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	75.9	61.4	58.6	73.6	60.5	57.1	69.5	58.8	27.1	62.7	25.9	60.9	24.8	59.5	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 61.2	54.7	49.8	DB	27.6	69.5	2.7	4.4	25.7	72.6	24.1	75.2	22.6	77.7	20.6	80.9	(4)
(5)			WB	24.9	61.9	2.5	2.8	23.2	63.9	21.7	65.5	20.3	67.1	18.5	69.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	41.2	40.3	40.8	43.4	46.7	50.9	55.0	56.2	53.5	48.8	44.8	42.4
	DBStd	6.54	3.34	3.92	3.57	3.34	3.51	3.15	3.70	3.20	3.24	3.33	3.58	4.05
	HDD50	1623	273	272	284	199	113	22	1	0	7	59	156	237
	HDD65	6552	738	692	749	647	567	423	311	273	344	501	605	702
	CDD50	547	0	0	0	1	11	49	157	193	113	22	1	0
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	25.5	31.4	29.7	27.3	24.0	21.8	21.4	19.8	20.6	24.3	27.5	28.5	29.5
Precipitation	PrecAvg	52.8	6.3	5.2	5.0	2.8	1.9	2.6	3.2	3.9	4.6	5.6	5.5	6.1
	PrecMax	65.0	9.9	10.1	9.3	4.8	3.3	5.2	6.0	6.4	7.5	8.5	8.7	9.4
	PrecMin	37.6	3.0	2.3	1.8	1.0	0.9	1.0	1.6	2.6	1.9	2.3	1.7	2.6
	PrecStd	6.5	1.9	2.1	1.8	0.9	0.6	0.9	0.9	1.0	1.3	1.6	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	48.6	48.5	53.9	61.1	65.8	71.2	68.3	64.1	55.8	52.2	50.3
		MCWB	47.4	47.5	46.3	46.6	52.6	56.6	61.0	59.4	57.5	52.2	50.2	49.2
	2%	DB	48.1	47.9	46.8	50.3	55.6	59.4	66.1	64.5	60.8	55.1	51.5	49.7
		MCWB	47.2	46.8	45.1	45.8	49.5	54.5	60.1	60.0	56.7	52.0	49.6	48.7
	5%	DB	46.7	46.5	46.5	48.4	52.2	55.8	62.2	62.4	59.0	53.8	50.2	48.4
		MCWB	45.3	45.3	44.8	45.6	48.4	52.3	58.7	59.5	55.7	51.1	48.5	47.3
	10%	DB	46.3	46.0	45.0	46.8	50.4	54.0	59.3	60.5	57.4	53.3	49.8	47.9
		MCWB	44.7	44.5	43.2	44.9	47.8	51.2	57.0	57.8	55.1	50.7	48.1	46.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	48.3	47.5	48.6	54.0	58.3	63.2	63.4	59.3	54.7	51.8	50.0
		MCDB	48.4	48.5	47.6	50.0	59.8	63.1	68.3	65.3	61.6	55.2	52.1	50.1
	2%	WB	46.9	46.6	46.3	47.3	51.3	54.9	61.0	61.5	57.6	53.5	50.3	48.5
		MCDB	47.2	46.9	46.6	48.4	53.9	58.7	64.4	63.5	59.4	54.2	50.6	48.6
	5%	WB	45.7	45.5	45.0	46.4	49.9	53.6	59.4	59.6	56.8	52.3	49.2	47.5
		MCDB	46.4	46.0	45.5	47.5	51.3	55.7	61.9	61.5	58.5	53.4	49.6	47.9
	10%	WB	44.4	44.0	43.9	45.5	48.5	52.5	57.6	58.2	55.7	51.3	48.0	46.2
		MCDB	45.5	44.9	44.8	46.7	49.9	54.4	59.5	59.9	57.3	52.6	48.7	47.1
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.1	4.8	4.7	4.5	4.1	4.1	4.1	4.1	4.0	4.4	4.7
		MCDBR	5.3	5.6	5.1	5.8	6.9	6.7	7.6	6.9	5.6	4.4	4.5	4.9
		MCWBR	6.4	6.6	5.7	4.4	4.5	4.5	4.6	4.7	4.3	4.6	5.6	6.4
	5% WB	MCDBR	5.1	5.5	4.5	4.8	5.4	5.9	6.2	5.8	4.8	4.0	4.5	4.8
		MCWBR	6.6	7.0	5.9	4.7	4.4	4.5	4.5	4.9	4.4	5.2	6.3	6.7
Clear-Sky Solar Irradiance	taub	0.334	0.355	0.368	0.382	0.370	0.363	0.381	0.372	0.365	0.347	0.321	0.385	
	taud	2.070	2.193	2.271	2.302	2.409	2.469	2.438	2.461	2.430	2.312	2.136	2.312	
	Ebn at Noon	111	190	235	257	271	275	266	259	237	193	116	70	
	Edh at Noon	14	23	29	34	33	31	32	29	25	20	13	9	
All-Sky Solar Radiation	RadAvg	62	220	583	1139	1591	1633	1477	1164	705	331	97	38	
	RadStd	8	30	72	104	109	178	124	102	56	42	13	3	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+1.35	N/A	N/A	N/A	N/A	-178	-256	N/A	N/A
Regional (24 neighbors)	N/A	+1.17	N/A	N/A	N/A	N/A	-149	-212	N/A	N/A

Nomenclature: See separate page